

CV & Publication List - Beck Strohmmer



Copenhagen



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RESEARCH FOCUS: Computational models of disease

EDUCATION

Apr 2022 PhD in Neuromorphic Engineering, University of Southern Denmark (SDU), Denmark
Jan 2019 MSc in Electronics Engineering, SDU, Denmark
Jun 2016 AP degree in Electronics Engineering, Erhvervsakademiet Lillebælt, Denmark
Dec 2003 BA in Physics and Mathematics, Bennington College, USA

POSITIONS

May 2026- Guest researcher, Dept. of Neuroscience, University of Copenhagen (KU)
May 2024-Apr 2026 Postdoc, Dept. of Neuroscience, University of Copenhagen (KU)
May 2022-Apr 2024 Postdoc, Dept. of Electrical and Photonics Engineering, DTU
Feb 2022-Apr 2022 Research assistant, Dept. of Electrical and Photonics Engineering, DTU

FUNDING

Sep 2023 Open call for Aage og Johanne Louis-Hansens Fond, "Modeling the temporal dynamics of neural circuit degeneration in ALS" (23-2B-14112) 1.263.150 DKK
Apr 2023 Lundbeck Foundation Investigator Network Seed Grant, "Temporal dynamics of neural circuit degeneration in ALS" (R426-2023-158) 150.000 DKK
Apr 2023 Lundbeck Foundation Investigator Network Seed Grant, "Dissecting the link between vision and freezing of gait in Parkinson's Disease" (R426-2023-104) 150.000 DKK

RESEARCH SUPERVISION

2025- Co-supervisor for PhD project, Finbar Haughian, KU & U of St. Andrews
2025-2026 Co-supervisor for Bachelor's Project, Angus Gray, KU & U of St. Andrews
2025-2026 Co-supervisor for Bachelor's Project, Beatrice Baldi, KU & U of St. Andrews
2022-2023 Co-supervisor for Master's Thesis, Ewa Elzbieta Rusiecka, DTU
2022 Co-supervisor for Master's Thesis, Maria Helena Neves, DTU
2022 Co-supervisor for Bachelor's Thesis, Jackii Sava Eleanor Alice Lily Jack Jim Seligman, DTU
2019-2020 Co-supervisor for Master's Thesis, Haukur Kristinsson, SDU

COMMISSIONS OF TRUST AND SCIENTIFIC ACTIVITIES

2026- Alumni Board Member of the Lundbeck Foundation Investigator Network, Denmark
2026- Grant Reviewer for My Name's Doddie Foundation, UK
2026- Member of the Neuroscience Academy of Denmark, Denmark
2026- IOP Publishing Associate Member, UK
2025- IOP Publishing Trusted Reviewer, UK
2025- Member of the The Euan MacDonald Centre for MND Research, UK
2025- Affiliate of the Pioneer Centre for Artificial Intelligence, Denmark
2024- Member of the Federation of European Neuroscience Societies, Denmark
2024-2025 Co-organizer of Lundbeck Foundation Investigator Network Neuroscience Winter School
2022-2026 Member of the Lundbeck Foundation Investigator Network, Denmark

SCIENTIFIC PRESENTATIONS

2026: Guest course lecture, U of St. Andrews; Departmental presentation, U of St. Andrews; 2025: Departmental presentation, Drexel Univ; Guest course lecture, KU; EMBO ABCD Meeting; Departmental presentation, U of St. Andrews; 2024: Departmental presentation, U of St. Andrews; Departmental presentation, Hvidovre Hospital; 2022: Departmental presentation, Drexel Univ; Guest course lecture, KU; Departmental presentation, U of Cologne; Guest course lecture, DTU; 2021: Departmental presentation, Italian Institute of Technology

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NATIONAL AND INTERNATIONAL RELATIONS

Comp. modeling Assoc. Prof. Jessica Ausborn, Dept. of Neurobiology & Anatomy, Drexel U, USA

ALS pathology Assoc. Prof. Ilary Allodi, School of Psychology and Neuroscience, U of St. Andrews, UK

TEACHING

2022 Lecturer, Interdisciplinary Bioengineering (Module 4 - Modeling the Brain and Neurorobotics), DTU

2022 Assis. Lecturer, Interdisciplinary Bioengineering (Module 4 - Modeling the Brain and Neurorobotics), DTU

2020 Teacher, Real-Time Systems (12-week course), SDU

2019 Teacher, Real-Time Systems (12-week course), SDU

2018 Assis. Teacher, Embedded Software Design (12-week course), SDU

OTHER

2025- Board member, KU LGBTQIA+ employee network

2021-2022 Head of organizing committee of CODE: Community for Openness and Diversity in Engineering at the Mærsk McKinney Møller Institute, University of Southern Denmark

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PUBLICATIONS (first author):

Strohmer, B., Grosh, K., Montañana-Rosell, R., Mora, S., Ausborn, J.* & Allodi, I.* (2025). Spinal circuit mechanisms constrain therapeutic windows for ALS intervention: A computational modeling study. *Neurobiology of Disease*, 107253.

Strohmer, B.*, Najarro, E.*, Ausborn, J., Berg, R.W., & Tolu, S. (2024). Sparse Firing in a Hybrid Central Pattern Generator for Spinal Motor Circuits. *Neural computation*, 36(5), 759-780.

Strohmer, B.*, Mantziaris, C.*, Kynigopoulos, D., Manoonpong, P., Larsen, L. B., & Büschges, A. (2022). Network architecture producing swing to stance transitions in an insect walking system. *Frontiers in Insect Science*.

Strohmer, B., Stagsted, R. K., Manoonpong, P., & Larsen, L. B. (2021). Integrating Non-Spiking Interneurons in Spiking Neural Networks. *Frontiers in Neuroscience*, 15, 184.

Strohmer, B., Manoonpong, P., & Larsen, L. B. (2020). Flexible Spiking CPGs for Online Manipulation during Hexapod Walking. *Frontiers in Neurorobotics*, 14, 41.

Strohmer, B., Bøgild, A., Sørensen, A. S., & Larsen, L. B. (2019). ROS-Enabled Hardware Framework for Experimental Robotics. In *2019 International Conference on ReConFigurable Computing and FPGAs (ReConFig)* (pp. 1-2). IEEE.

PUBLICATIONS (co-author):

Kim, K., Sonogo, V., Strohmer, B., Chang, B., Liu, X., Yonza, A. K. I., Kucharz, K., Asiminas, A., Lind, B., Tao, L., Zhang, X., Fried, S., Han, A., & Cai, C. (2025). Single-Cell and Population-Level Neuromodulation Dynamics in Dual-Electrode Intracortical Stimulation. *bioRxiv*, 2025-06. (In review)

Tolu, S., Strohmer, B., & Zahra, O. (2023). Perspective on investigation of neurodegenerative diseases with neurorobotics approaches. *Neuromorphic Computing and Engineering*.

Thor, M., Strohmer, B., & Manoonpong, P. (2021). Locomotion control with frequency and motor pattern adaptations. *Frontiers in Neural Circuits*, 135.

Larsen, L. B., Stagsted, R., Strohmer, B., & Christensen, A. L. (2021). CloudBrain: Online neural computation in the cloud. *bioRxiv*.

PUBLICATIONS (software):

Linssen, C., Babu, P., Eppler, J., Ewert, L., Blaszczyk, K., Wybo, W., Strohmer, B., Rumpe, B., & Morrison, A. (2023). NESTML 7.0.0. Zenodo. doi:10.5281/zenodo.10125416.

*Authors contributed equally to this work.